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Spotlight

Yet uninfected? Resolving cell states of plants under pathogen attack

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Although we have made significant strides in unraveling plant responses to pathogen attacks at the tissue or major cell type scale, a comprehensive understanding of individual cell responses still needs to be achieved. Addressing this gap, Zhu et al. employed single-cell transcriptome analysis to unveil the heterogeneous responses of plant cells when confronted with bacterial pathogens.

The plant biology community has embraced single-cell transcriptome (single-cell RNA sequencing [scRNA-seq]) technologies, allowing researchers to view plant tissue with the compound lens of genomics and cell biology.[1] In addition to enabling granular and comprehensive analyses of known cell types, scRNA-seq can reveal different states of each cell type that are influenced by cell-intrinsic and cell-extrinsic factors. The cell cycle represents a cell-intrinsic factor, whereas environmental stimuli contribute as cell-extrinsic factors. A particularly heterogeneous environmental stimulus is pathogen infection because their distribution is non-uniform and individual cells of pathogens respond to plants independently using genetic programs co-evolved with the hosts. For instance, distinct fungal pathogen *Magnaporthe oryzae* infection stages were observed in a single rice leaf.[2] Plant cell responses can also be highly variable depending on their spatial relationships with pathogen cells and activity. Moreover, different plant cell types possess the distinct potential to respond to microbial signals, as previously shown by transcriptome analyses of sorted cell populations.[3]

Key questions in plant-pathogen interactions that can be addressed through scRNA-seq analyses include: (1) how does the spatial heterogeneity of infection contribute to molecular heterogeneity in individual cells? (2) How do known immune and susceptible pathways function in individual cells? (3) Can single-cell analyses identify specific cell states in pathogen-infected plants and reveal genes and pathways that operate in distinct cell states, which may have been overlooked in previous bulk tissue analyses?

Zhu et al.[4] tackled these questions by employing a model plant pathosystem, where *Arabidopsis thaliana* leaves are challenged by the bacterial pathogen *Pseudomonas syringae* pv. tomato DC3000 (*Pst* DC3000). *Pst* DC3000 uses effector molecules and toxins to subvert the plant immune system, proliferating in the leaf intercellular space (apoplast) and causing disease symptoms in plants.[5] Observation of fluorescently labeled *Pst* DC3000 revealed a heterogeneous distribution of the pathogen cells during infection in *A. thaliana*. For scRNA-seq analysis, the authors isolated cells (protoplasts) by digesting the cell wall of leaves 24 h after infection (syringe infiltration). Because *Pst* DC3000 mainly proliferates in the apoplast surrounded by mesophyll cells, the authors employed a protoplast method known as the "Tape-Arabidopsis Sandwich method" that enriches this cell type. The authors also performed bulk RNA-seq with and without the protoplasting process and identified 7,548 genes induced by protoplast treatment; these genes were removed from scRNA-seq analyses.

As expected, most cells (~94.7% of 11,895 cells) profiled by scRNA-seq were annotated as mesophyll cells, but these cells showed diverse transcriptome patterns in pathogen-infected leaves. The authors identified five mesophyll subpopulations (clusters) affected by pathogen infection. Gene Ontology analysis identified distinct clusters enriched with genes responsive to different plant hormone pathways, salicylic acid (SA) and jasmonic acid (JA), as well as an intermediate cluster. Based on known genes involved in immunity and susceptibility against the bacterial pathogen, the authors confirmed that the clusters with SA and JA responses were enriched with immunity and susceptibility genes, respectively. Together, the scRNA-seq data identified distinct cell states within a cell type spanning from immunity to susceptibility states influenced by pathogen infection.

The authors then sought to understand how cells transition between identified cell states during pathogen infection. They employed pseudotime analysis, a method commonly applied for inferring developmental trajectories by aligning cells as trajectories based on their gene expression patterns. Pseudotime analysis applied to mesophyll cells revealed a mostly linear trajectory that progressed through non-pathogen-responsive clusters, followed by immune clusters, an intermediate cluster, and ended in the susceptible clusters. Furthermore, the authors identified genes strongly expressed at distinct phases of the disease progression trajectory, providing valuable information for further characterizing each cell state.

The authors validated gene expression dynamics predicted by pseudotime analysis using transgenic reporter lines that visualize promoter activity associated with selected genes. Reporter plants were